

# Stochastic Optimization Practice

## Food Bank Inventory Management

2024-2025

### 1 Introduction

A food bank is an emergency feeding organization that provides hunger relief to families living in poverty. We consider a food bank with two warehouses, denoted as  $\mathcal{I} = \{1, 2\}$ , that prepare food weekly for six districts, represented by  $\mathcal{J} = \{1, 2, \dots, 6\}$ . The food demand, denoted by the random variable  $\xi_{d,j}$ , exhibits both spatial dependency on  $j \in \mathcal{J}$  and temporal dependency on the day  $d \in \mathcal{D} = \{1, 2, \dots, 7\}$  (Monday to Sunday) of a given week. The locations of the warehouses and districts are depicted in Figure. 1.

The decision-maker must determine the inventory amount for each warehouse, denoted by  $x_i$ , on Sunday night. Starting from Monday, the two warehouses supply food  $y_{d,j}$  to each district on a daily basis. Specifically:

- Warehouse 1 serves districts 1–3, while warehouse 2 serves districts 4–6.
- There is an opportunity on Friday night for warehouse 1 (or 2) to transfer food to warehouse 2 (or 1) based on consumption needs, denoted as  $m$ , to support the other warehouse.

The objective is to optimize the weekly ordering amount  $x_i$  and make daily planning decisions while satisfying operational constraints. The key requirements are as follows:

- The food bank must determine  $x, y$ , and  $m$  (positive if from 1 to 2) each Sunday.
- Each meal incurs an ordering cost  $c_i^o$ , which differs for each warehouse.
- Any leftover food at the end of the week will spoil, and the food bank must pay a waste disposal cost  $c^w$  per meal.
- Regulations recommend that at least 95% of the weekly demand for each district be satisfied.
- Since  $y_{d,j}$  is planned in advance, it may not match  $\xi_{d,j}$  exactly. The food bank follows these operational rules:

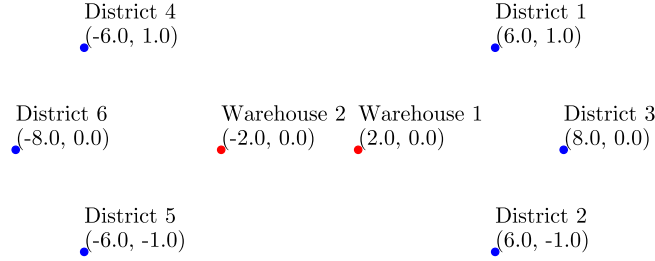


Figure 1: Location of warehouses and districts.

- If excess food is supplied to a district, it is returned to the warehouse and must be transported back to the same district the following day before redistribution.
- Warehouses cannot supply additional food beyond  $y_{d,j}$  plus the left-over from yesterday to districts on a given day.
- However, within each warehouse’s service group (i.e., districts  $\{1, 2, 3\}$  and  $\{4, 5, 6\}$ ), food can be reallocated among districts with an emergency transportation fee  $c_{j,j'}^e$  per meal.
- A nightly holding cost  $c^h$  per meal applies from Sunday through Saturday.
- The daily transportation cost from warehouse  $i$  to district  $j$  is  $c_{i,j}^f$  per meal. For simplicity, return transport costs are ignored.
- The Friday transfer amount  $m$  (from warehouse 1 to 2 if positive) is cost-free but constrained:
  - If  $m > 0$ , it must be less than 15% of the initial inventory  $x_1$ .
  - If  $m < 0$ , it must be less than 20% of the initial inventory  $x_2$ .

## 2 Tasks

This practice consists of two tasks, one mandatory and one optional.

### 2.1 Task 1: Static Case (Mandatory 100%)

You are provided with historical weekly demand data covering five years. Each year consists of 52 weeks, starting from Monday, January 2, 2012. The data files are named  $j$ -**training.npy**, where  $j$  corresponds to a district, and each

file has the shape [5, 52, 7]. Your task is to formulate a stochastic programming model using Sample Average Approximation (SAA) and solve it. Specifically:

- Determine the inventory management plan, including  $x, y$ , and  $m$ . These decisions should be deterministic, i.e., identical for each week.
- An advanced solution strategy (Bender’s Decomposition, Progressive Hedging, etc.) may be considered instead of solving the full-scale model directly.
- Summarize your modeling approach, solution strategy, and results in the provided .ipynb file. Your work will be evaluated by
  - (70%) Correctness of the model formulation and decision quality (cost, whether regulation is satisfied, the food waste rate).
  - (30%) Solution strategy and computational efficiency.
- Relevant cost parameters are also given in the .ipynb file. We recommend cvxpy as the modeling language.

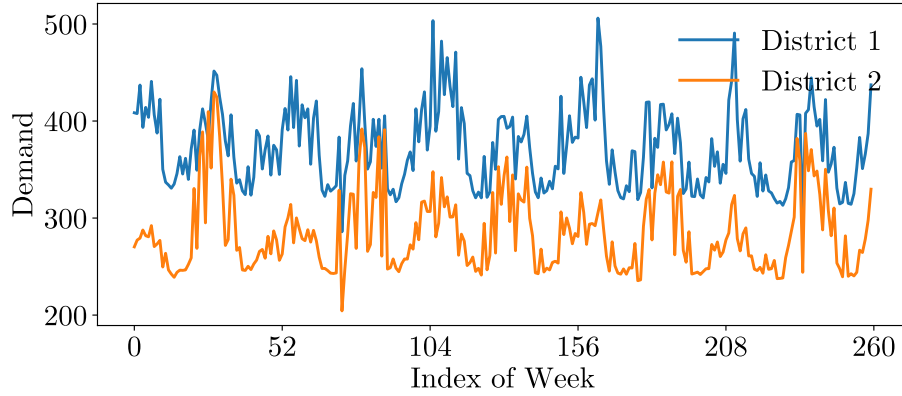


Figure 2: Monday demand for districts 1 and 2 over time.

## 2.2 Task 2: Dynamic Case (Optional Bonus)

Figure. 2 illustrates the Monday demand for districts 1 and 2 across all weeks in the dataset. The demand clearly exhibits temporal dependency and seasonality. Consequently, the static SAA solution may be overly conservative during low-demand weeks, leading to excessive waste. To address this, a dynamic policy is required, where decisions are tailored to each week’s characteristics.

Your task is to construct a policy that utilizes relevant temporal information to optimize decisions for specific weeks. Implement your policy to determine  $x, y, m$  for the 10th, 25th, and 43rd weeks of the 6th year (of which the demand data are hidden). Your decisions will be evaluated using the same metrics as

in Task 1. Please document and your approach to construct the policy in the same file.

To achieve this task, you may need extra reading materials regarding contextual decision-making. Please refer to [1] and [2] for a broader understanding. To guarantee the performance of your policy on unseen data, you may need to improve the robustness of your policy. For instance, if your policy contains some parameters, you may consider splitting your training data properly for cross-validation.

### 3 Submission

You will find a .rar file from the Edunao platform containing a directory of data, a directory used for saving the results, and a notebook used to formulate models and algorithms. Once you finish the task, keep the running log of the notebook and save required results. Later on, zip up the project and send it to me by email. Name the subject of emails as [SO-2025-P2-YOUR NAME] (Only the sender, add other names in the notebook). For those who cannot download the .rar file, please try this link:

<https://nextcloud.centralesupelec.fr/s/8SyX8JF2Mecj7B5>.

#### **Bibliography**

[1] Sadana, U., Chenreddy, A., Delage, E., Forel, A., Frejinger, E., & Vidal, T. (2024). A survey of contextual optimization methods for decision-making under uncertainty. *European Journal of Operational Research*, 320(2), 271-289.

[2] Bertsimas, D., & Kallus, N. (2020). From predictive to prescriptive analytics. *Management Science*, 66(3), 1025-1044.